Table 6.2: notation of the leak codes

code	meaning	evidence
M	identical variant row in training (same protein family, identical mutation set, including reverse structures)	verified
m	every mutation in the row appears in training in the same protein, but not this combination (e.g. a test double whose two singles were trained)	verified
P	same protein (≥95 % identity over ≥90 % shorter-sequence coverage, or ≥90 % identity over ≥50 %) in training, no mutation match	verified
H	strong homolog in training: MMseqs2 E ≤ 1e-3 over ≥50 % of the test protein (plus Foldseek E ≤ 1e-3 or qTM > 0.70 over ≥50 % on Domainome and ProteinGym)	verified
h	weak/ambiguous: MMseqs2 E ≤ 1e-2 or ≥30 % identity over ≥30 % of the test protein (plus Foldseek E ≤ 1e-2 over ≥30 % on Domainome and ProteinGym)	measured
D	documented training on the same source dataset; rows not reconstructable	documented
–	no supervised ΔΔG training overlap (includes physics-based and unsupervised pretraining-only models)	verified / documented
V	the rows were used for validation, model selection or calibration (Rosetta's energy-function tuning)	split file / paper
#	the number is not a test result (the dataset is the training set)	verified
?	ceiling, not verified: inferred from a published protein-level split because the training rows are not public (IFUM M?, V?)	inferred
n.a.	not audited: no statement about overlap is made, and the cell is not eligible for bold	–

Table 6.3: training inventories

inventory	training data	source	evidence
ESM-MSR (released), ESM-MSR (devel), σ sweep, WT/MT heads	all substitutions of the 172 capped-train libraries (229,721 rows)	splits_oct06_capped.pkl + Tsuboyama2023_Dataset2_Dataset3_20230416.csv	verified
MutateEverything st1	the 172 capped-train libraries: 154,358 valid singles (150,815 wild-type-backbone + 3,543 mutant-background)	capped split; ME README recipe, stage 1 (20 epochs)	verified
MutateEverything st2	the same singles plus all 53,188 measured doubles (207,546 rows); stage 2 (100 epochs) initialised from the same seed's stage 1	capped split; ME README recipe	verified
ThermoMPNN (additive)	the same 154,358 singles	capped split; train_single.yaml	verified
ThermoMPNN-D	134 of the 172 capped-train libraries: the 53,188 measured doubles (52 libraries) plus 118,963 Rosetta-augmented doubles (82 additional natural libraries)	capped split; train_epistatic.yaml (October audit; the retrained-baseline inventories were retired from the pipeline on 2026-11-12, LEAKAGE_AUDIT_OCT2026.md §11)	verified
SPURS (retrained)	the 172 capped-train libraries: 154,358 valid singles (150,815 wild-type-backbone + 3,543 mutant-background)	capped split; patched SPURS training code, three seeds	verified
SPURS multi (retrained)	the same singles (through its same-seed SPURS initialisation) and the 53,188 measured doubles	capped split; patched SPURS training code, three seeds	verified
EsmTherm	K50_dG_Dataset1_Dataset2: 47 test-only domains; the other 414 split 80/10/10 per sequence (527,785 sequences with split labels)	S2 Spreadsheet pcbi.1012248.s006.csv (PLoS Comput Biol 2024) + wildtype_split.csv	verified (exact-sequence join)
IFUM	Megascale Dataset2, MMseqs2-30 % cluster split: 322 train domains, 86 in an unseparated val/test block	IFUM published dataset_splits.pdf	protein-level (M? / V? ceilings from the public Dataset2 rows of those domains)
DDGemb	S2450 = S2648 − 18 sequences/198 mutations >25 % identity to S669; 2,450 mutations / 115 UniProt, doubled with reverse variants	ddgemb.biocomp.unibo.it S2450.tsv + S2450.fasta	verified (sequence-aligned, forward + reverse)
DDMut	single-mutant model: S9028 = S2648 + FireProtDB, 4,514 forward + 4,514 reverse; deployed multi-mutant model: SM1218 training + SM444 blind test (831 + 831 reverse)	biosig.lab.uq.edu.au S9028.csv, DD_SM1218.csv, DD_SM444.csv	verified (forward + reverse, union of both models)
GeoDDG-3D	S8754 = 8,754 single mutations / 301 proteins (ProThermDB + ThermoMutDB); encoder pretrained as GeoFitness on 74 DMS fitness sets (52 MaveDB + 22 DeepSequence)	GeoStab S8754.csv; data/dms/dms.csv	verified (construct-aligned); GeoFitness protein-level
ESM3ΔG	960,216-sequence MGnify Stability fine-tune (525,127 WT + 435,089 point mutants); Megascale is its held-out test	Zenodo 19411306 mgnify_training_index.csv	verified (0 exact sequences)
ProSST	18.8 M AlphaFoldDB structures (masked-LM); no ΔΔG fine-tune (–)	ProSST paper	documented only
ProteinMPNN	PDB assemblies deposited before 2021-08-02; no mutation-level supervision (–)	ProteinMPNN paper	documented only
ESM3 (small/medium/large, and the σ = 0 control)	UniRef/MGnify/JGI + PDB pre-2020-05-01 + AFDB v4; no mutation-level supervision (–)	ESM3 paper	documented only
Rosetta Cartesian ΔΔG (2019, beta_nov16_cart)	no per-variant training; 20 reference energies fit with Kellogg-2011 ProTherm ΔΔG ranks and Rocklin-2017 design stabilities in the objective, REU → kcal/mol 1/2.94 fit on Kellogg	Rosetta docs Updates-beta-nov16; Park et al. 2016 JCTC SI	calibration (V)
ESM-MSR-small (March reference)	the 120 train libraries of the legacy hyperopt split; selected on its 26 val libraries	data/hyperopt_splits.pkl	verified

Table 6.5: what the audit finds

finding	size	affected rows
23 ProteinGym assays are fully inside the capped train split (+ 3 strong 3D structural homologs)	23/217 M + 3/217 H DMS, all Stability	45,623 M + 2 m + 4,607 H / 2,465,767 rows
ME stage 1/2, ThermoMPNN and retrained SPURS on those assays	singles (24,252 exact) + constituent (21,331) for ME/SPURS; 24,252 exact for TMPNN	same 23 M + 3 H DMS
Two Domainome domains (O95155_PF04564_1229 = 3L1X, Q9H161_PF00046_216 = 2M0C) match capped train at 100 % identity (+ 60 strong 2D/3D homolog domains)	2 M + 2 P + 60 H of 522 domains (458 low-id clean of M/P/H)	2,023 exact + 2,288 P (0.8 %) + 58,422 H rows (10.9 %); h.03 on the 458-domain subset
EsmTherm's 80/10/10 is per sequence, not per domain: 26,789 capped-test rows are identical training variants, verified against its published sequences	51.4 % identical (M.51) + 34 % same protein; 3,155 rows only in its validation split (V.06)	M.51 P.34 H.04 h.03 V.06
IFUM's train domains cover 21/33 test libraries (including mutant backgrounds)	24,200 rows are Dataset2 rows of its train domains (M?.46, a ceiling); 24,608 more sit in its val/test block (V?.47)	M?.46 P.02 H.07 h.06 V?.47
ESM3ΔG is zero-shot on Megascale	0 exact sequences anywhere in the benchmark	40.7 % strong (8/33 libs) + 50.3 % weak (21/33) homologs
DDGemb = S2450 forward + reverse (sequence-aligned S2450.tsv + S2450.fasta)	91.7 % of S2648; 63.4 % of Q3421; 43.6 % of S8754; 38.3 % of K3822; 33.0 % of K2369; 59.6 % of Ssym; m.48/m.54 on PTMUL/PTMUL-D; 0 % on S783, S669, S461	verified exact / constituent
DDMut = S9028 + SM1218 + SM444 (forward + reverse structures)	98.2 % of S2648; 88.3 % of Ssym; 83.5 % of Q3421; 74.3 % of K2369; 72.2 % of K3822; 69.5 % of S8754; 60.9 % of S783; 79.5 % + 11.3 % constituent / 95.0 % of PTMUL/PTMUL-D; 10.0 %/14.5 % of S669/S461 (plus 19 %/24 % same protein)	verified exact
GeoDDG = S8754 (construct-aligned)	S8754 is its training set (#); 100 % of S2648; 97.9 % of Q3421; 91.2 % of K2369; 89.1 % of K3822; 88.4 % of S783; 50.0 % of Ssym; m.73/m.78 on PTMUL/PTMUL-D; 1.2 % of S669; GeoFitness pretraining proteins: DLG4 abundance/binding (P1), 11 % of S461, 7 % of S669	verified exact / constituent; GeoFitness protein-level
the "six clean sets" claim	only S461 has zero identical mutations across all full-coverage models	DDMut has M.10/M.15 on S669/S461; GeoDDG has M.01 P.07 on S669 and P.11 on S461
Rosetta's beta_nov16 tuning data	Kellogg 2011 mutations in 4–28 % of the classical sets (Q3421 V.28, S783 V.22, Ssym V.19, K3822 V.18, S2648 V.16); Rocklin-2017 designs = 7/33 capped-test and 3/23 val scaffolds	V.12 test, V.07 val, V.002 ProteinGym (protein level)
ESM-MSR-small (March reference) homology was never computed	strong homologs on 9 of 12 external sets, ≥ 0.5 % on S2648, S8754, Q3421, PTMUL, PTMUL-D	H.02, H.01, H.01, H.05, H.03